

Exercise 10 - Master Theorem Problems

Create a file named '`master.txt`'. Within the file, for each recurrence function below, find its runtime function by proving which Master Theorem case it represents.

Problems:

1. $T(n) = T(n/2) + O(1)$
2. $T(n) = 2T(n/2) + O(n)$
3. $T(n) = 8T(n/2) + O(n^2)$
4. $T(n) = 2T(n/2) + O(n^2)$
5. $T(n) = T(n/2) + O(n)$
6. $T(n) = 2T(n/2) + O(n \lg(n))$
7. $T(n) = 4T(n/2) + O(n)$
8. $T(n) = 3T(n/4) + O(n)$
9. $T(n) = 4T(n/4) + O(n)$
10. $T(n) = 7T(n/2) + O(n^2)$